The complete string C-group counts for S_{10}

Human Compression project

We recover the S_{10} row of Table 1 in the fixed edition of Leemans' survey [1]. Its definition identifies generating sets under group automorphisms. As in the acquired eight-point calculation [3], the connected noncommutation path shows that this is exactly the equivalence generated by simultaneous conjugacy and reversal of the ordered marking.

Theorem 1. *The complete counts for S_{10} are*

rank	3	4	5	6	7	8	9
ordered classes	809	403	99	26	13	2	1
self-dual classes	17	3	5	0	1	0	1
classes with duality identified	413	203	52	13	7	1	1.

There are no representations of larger rank.

Proof. Apply the complete permutation enumeration of [2], without altering its mathematical engine. The first mark is a product of e disjoint transpositions, where $1 \le e \le 5$. These are exactly the five involution classes of S_{10}. Each of the five independent initial branches finishes through rank nine. The retained Whiston independent-generator bound excludes every larger rank. Thus the acquired prefix-orbit theorem gives a complete list, with no ordered point-conjugacy repetitions. Its rejection of commuting adjacent marks is valid here because S_{10} has no nontrivial direct-product decomposition. The resulting list contains 1353 tuples, with the first-row distribution above.

A separate actual-group verification checks the order 10! of each generated subgroup, all marked involutions and distant commutations, and all pairs of subsets of every marking. There are 969 024 checks of

$$\langle \rho_i : i \in I \rangle \cap \langle \rho_j : j \in J \rangle = \langle \rho_k : k \in I \cap J \rangle.$$

These checks certify the positive objects; completeness follows from the finished enumeration and its acquired proof.

The independent coloured-traversal invariant of [3], now minimized over all ten roots, verifies all ordered class distinctions. It also identifies the unique retained class of each reversed tuple. The resulting permutation is an involution, with fixed-point counts $17, 3, 5, 0, 1, 0, 1$. Its orbits therefore number half the sum of the first two rows, which gives the last row. All automorphisms of S_{10} are inner; the finite verifier also checks $|\operatorname{Aut}(S_{10})| = 10!$. The noncommutation graph of the marked generators is a path, so its only possible permutations under a generating-set isomorphism are the identity and reversal. This pays exactly the source convention. □

Verification and attribution

The source already reports these values. This component supplies a complete execution of the acquired method and independently checks the actual subgroup and reversal interfaces. It introduces no new enumeration theorem and makes no conjecture or priority claim. The full literal records and terminal receipts are retained. All five branches use the same frozen engine; the launcher change merely permits this additional fixed source row, with a separate bounded deadline for each initial class. A capped branch would establish no classification, and no capped result is included here.

The checker is the literal degree-ten specialization of the accepted eight-point wrapper. Its complete difference is supplied. The independent array canonicalization and subgroup validator have the same precise specialization. Damaged and truncated certificates must reject, and Python optimization is rejected before output work. The default replay repeats all five initial branches and the complete subgroup and literal-array checks. Correctness of GAP's exact permutation algorithms remains explicit computational infrastructure.

The values in ranks seven through nine were already acquired from the stable counting theorem. The four cells in ranks three through six are the additional recovered entries. The remaining survey tables, whole-original admission and subsequent compression audit are separate obligations.

References

[1] D. Leemans, *String C-group representations of almost simple groups: a survey*, arXiv:1910.08843v2, Table 1, physical page 6.

[2] Human Compression project, *Complete permutation searches and the six-point row*, accepted finite-method component, with its exact independent review and root adoption.

[3] Human Compression project, *The complete string C-group counts for* S_8, accepted component and distinct root review.